

Version 5/17

Mingxiao Li

Department of Industrial Engineering

2026 年 7 月 8 日

Notation

We summarize all mathematical notation organized by functional modules. Symbols introduced only in a specific context are listed within that module; symbols used globally are placed in the first relevant module.

1. System primitives: sets, grids, and basic parameters

- t, τ —Continuous time variables.
- T —Planning horizon length for a single swap routing tour.
- i, j —Grid (node) indices; 0 denotes the depot.
- $\mathcal{J}$ —Set of all grids; $\mathcal{N} = \mathcal{J} \cup \{0\}$.
- $\delta^+(j) = \{(j, k) : k \in \mathcal{J}, k \neq j\}$, $\delta^-(j) = \{(k, j) : k \in \mathcal{J}, k \neq j\}$ —Outgoing / incoming physical arcs of grid j .
- a_{ij} —Travel time from grid i to j (Haversine distance at constant speed 30 km/h).
- C —Battery carrying capacity of the service vehicle per tour.
- c —Per-battery swap service time.
- λ_j —User departure (demand) rate at grid j (Poisson, per unit time); λ : area-level aggregate.
- ρ_j —Bike arrival (return) rate at grid j (Poisson, per unit time); ρ : area-level aggregate.
- $n_{j,k}$ —Initial count of bikes in battery state k at grid j at $t = 0$.
- $N_{j,0} = \sum_{k=1}^2 n_{j,k}$ —Initial serviceable inventory at grid j (soon + normal).

2. Battery states

- $\mathcal{K} = \{\text{low, soon, normal}\}$ —Discrete battery-state categories: low ($< 20\%$, unavailable), soon ($20\%–50\%$, 80% service weight), normal ($> 50\%$, full service).
- $k \in \mathcal{K}$ or $k = 1, 2, 3$ —State index. In fluid ODEs only $k = 1, 2$ (soon, normal) are tracked; the low state is a static reservoir convertible only at swap moments.

3. Global and local state distributions

- Θ_k^{global} —Global (area-wide) proportion of bikes in state k , used to approximate the composition of arriving bikes at any grid (Assumption 2).
- $\Theta_{j,k}^{\text{grid}}(t) = N_{j,k}(t)/N_j(t)$ —Local proportion of state k at grid j at time t , where $N_j(t) = \sum_{k=1}^2 N_{j,k}(t)$.

4. Inventory dynamics (fluid ODE system)

- $N_{j,k}(t)$ —Number of bikes in state k at grid j at time t (continuous fluid).
- $N_j(t) = \sum_{k=1}^2 N_{j,k}(t)$ —Serviceable inventory at grid j .
- $N_k(t) = \sum_j N_{j,k}(t)$ —Aggregate serviceable bikes in state k across all grids.
- $N(t) = \sum_{k=1}^2 N_k(t)$ —Total serviceable inventory (soon + normal).
- $N_{\text{serv},j}(t) = N_{j,\text{soon}}(t) + N_{j,\text{normal}}(t)$ —Serviceable bikes at grid j (equivalent to $N_j(t)$).
- $N_{\text{in}}(t) \sim \text{Poisson}(\rho_j t)$, $N_{\text{out}}(t) \sim \text{Poisson}(\lambda_j t)$ —Cumulative arrival / departure counts over $[0, t]$ (expectation used in fluid limit).

5. Utility evaluation: stock-out, service value, and swap mechanics

- t_0 —Stock-out time under natural dynamics, defined by $N_{\text{serv},j}(t_0) = 0$.
- $\tilde{t}_0$ —Stock-out time after a swap intervention at u .
- $\tau_0 = \min(t_0, T)$, $\tilde{\tau}_0 = \min(\tilde{t}_0, T)$ —Effective service upper bounds (before / after swap).
- $S_j(u_j)$ —Baseline net service value collected from arrival u_j to τ_0 , measuring quality-weighted net outflow of service.
- $\tilde{S}_j(u_j, y_j)$ —Post-swap service value from u_j to $\tilde{\tau}_0$, after swapping y_j batteries.
- $U_j(u_j, y_j) = \tilde{S}_j - S_j$ —Marginal utility of a swap operation (the quantity maximized by routing).
- $\tilde{N}_{j,0} = N_j(u) + \min\{y_j, N_{j,\text{low}}(u)\}$ —Total post-swap serviceable inventory.

- $\tilde{n}_{j,k}$ —Post-swap count of state k at the swap moment (piecewise function of y_j , $N_{j,\text{low}}$, $N_{j,\text{soon}}$; see Eq. (4.7)).
- $\tilde{N}_{j,k}(t)$ —Post-swap inventory trajectory for $t > u$, governed by the same ODE form with shifted initial condition $\tilde{n}_{j,k}$ and time origin u .
- $\tilde{\Theta}_{j,k}^{\text{grid}}(t) = \tilde{N}_{j,k}(t)/\tilde{N}_j(t)$ —Post-swap state proportion.
- $\Delta_{\text{swap}} = \min\{y_j, N_{j,\text{low}}\} + 0.2 \max\{0, y_j - N_{j,\text{low}}\}$ —Net service-quality increment injected at the swap instant (closed-form; priority: low before soon, soon at 20% marginal gain).

6. Routing optimization: classical VRP decision variables

- $x_{ij} \in \{0, 1\}$ —Binary routing arc: 1 if vehicle travels directly from i to j .
- $v_j \in \{0, 1\}$ —Visit indicator: 1 if grid j is serviced during the tour.
- $y_j \in \mathbb{Z}^+$ —Number of batteries swapped at grid j .
- $u_j \geq 0$ —Vehicle arrival time at grid j (MTZ temporal tracking variable).
- M —Big-M constant in MTZ constraints (sufficiently large, e.g., $M = T + \max_{i,j} a_{ij} + c \cdot C$).

7. Utility tensor and discretization (MIP interface)

- P —Number of time discretization segments; $\mathcal{S} = \{0, 1, \dots, P\}$ is the set of breakpoint indices.
- τ_s —Physical time at discrete breakpoint $s \in \mathcal{S}$, with $\tau_0 = 0$, $\tau_P = T$.
- $\mathcal{Y} = \{0, 1, \dots, C\}$ —Feasible swap-quantity set.
- $\Omega_j(y, \tau_s)$ —Pre-computed utility tensor: marginal utility of swapping y batteries at grid j upon arrival at τ_s , computed offline via the closed-form U_j (Section 4).
- $\lambda_{j,y,s}$ —SOS2 interpolation weight variables mapping continuous arrival time to discrete breakpoints (used only in Scheme A; eliminated in STGraph).
- $\delta_1 \geq \delta_2 \geq \dots \geq \delta_P$ —Binary delta-staircase variables for arrival-time encoding (Scheme B; eliminated in STGraph).
- $f_{ij} \leq C \cdot x_{ij}$ —Single-commodity flow on arc (i, j) for tightening LP coupling (Scheme B; eliminated in STGraph).

8. Space-Time Graph (STGraph)

- $\mathcal{T} = \{0, 1, \dots, P\}$ —Discrete time-step index set (synonymous with $\mathcal{S}$; used when emphasizing DAG topology).
- (i, s) —Space-time node: vehicle is at physical node i at time τ_s .
- $e = (i^e, s_1^e, j^e, s_2^e, y^e)$ —Space-time arc tuple: depart i at τ_{s_1} , swap y^e batteries, arrive at j at τ_{s_2} .
- $\mathcal{E}$ —Set of all feasible space-time arcs (enumerated offline via right-rounding and KNN filtering).
- $x_e \in \{0, 1\}$ —Binary arc-activation variable for $e \in \mathcal{E}$.
- $\delta_{ST}^+(j, s), \delta_{ST}^-(j, s)$ —Outgoing / incoming space-time arcs of node (j, s) .
- K —Number of nearest spatial neighbors retained after KNN sparsification (default $K = 50$).

9. Operators

- $[x]_+ = \max(0, x)$ —Positive-part operator.
- $(\cdot)'$ or $\frac{d(\cdot)}{dt}$ —Time derivative.

1 Introduction

The dynamic scheduling of battery-swapping vehicles for shared micro-mobility networks represents a highly complex intersection of predictive data science, stochastic queuing theory, and combinatorial optimization. Unlike traditional routing models that seek to minimize distance or cost under deterministic assumptions, our proposed formulation seeks to maximize an energy-conversion utility derived from continuous-time fluid approximations of stochastic bike fleets.

2 Related Work and Literature Review

To establish the theoretical foundation for our approach, we synthesize four primary streams of literature: spatio-temporal demand forecasting, mean-field approximations for mobility networks, repositioning and routing in shared systems, and the emerging “Predict-then-Optimize” paradigm.

2.1 Spatio-Temporal Demand Forecasting in Urban Mobility

The effectiveness of any time-dependent mobility service inherently depends on the accurate estimation of underlying demand rates (i.e., user departure rate λ_j and return rate ρ_j). Traditional statistical time-series models often fail to capture the complex, non-linear spatio-temporal dependencies present in modern mobility networks. Consequently, machine learning—specifically tree-based ensemble methods—has become the industry standard for large-scale tabular forecasting. [5] revolutionized this space with XGBoost, providing a scalable end-to-end tree boosting system, further advanced by LightGBM [12] with histogram-based algorithms and Gradient-based One-Side Sampling (GOSS). To handle high-cardinality categorical identifiers (e.g., H3 hexagonal grids), CatBoost [14] introduced ordered boosting and target statistics, effectively preventing target leakage and dimensionality explosion.

In parallel, the deep learning community has proposed advanced architectures to explicitly model spatial spillover effects. [20] pioneered the Spatio-Temporal Graph Convolutional Network (STGCN), while [13] proposed Diffusion Convolutional Recurrent Neural Networks (DCRNN) to model traffic as a stochastic diffusion process on directed graphs. Further extending this, [10] developed multi-graph convolution networks for ride-hailing demand, and [21] introduced GMAN with graph multi-attention mechanisms. Despite the theoretical superiority of deep learning in modeling graph structures, tree-based models like CatBoost remain highly advantageous when preparing parameters for operations research applications: they produce lower-variance and more stable predictive parameters, which is critical to preventing severe oscillation and non-convergence in downstream Mixed-Integer Programming (MIP) solvers.

2.2 Mean-Field Approximations and Stochastic Fluid Limits

Modeling the exact state transitions of a large-scale bike-sharing network using discrete Markov chains inevitably triggers the curse of dimensionality. The foundational mathematical justification for circumventing this was provided by [2], who proved that as the number of interacting components approaches infinity, the stochastic discrete states converge to a deterministic fluid limit governed by Ordinary Differential Equations (ODEs). [9] further validated that expected values estimated via mean-field approximations are $O(1/N)$ -accurate, even for finite-sized systems.

In the specific context of shared mobility, [8] applied mean-field approximations to homogeneous bike-sharing systems, proving that complex local queuing dynamics can be effectively decoupled and analyzed through continuous equations, and revealed the critical role of “power of two choices” incentive mechanisms. In the ride-hailing domain, [4] utilized steady-state fluid limits of closed queueing networks to establish asymptotic optimal utility bounds for empty-car routing. Furthermore, [18] and [1] respectively applied fluid approximations and mean-field dynamics of discrete-time Markov decision processes to provide approximately optimal solutions for dynamic pricing and inventory management in micromobility systems.

Inspired by this framework, our model treats the inventory of serviceable e-bikes and their

aggregate battery energy as continuous fluid reservoirs. This approximation facilitates closed-form ODE solutions for post-swap energy decay, translating intricate queuing behaviors into deterministic, continuous parameters suitable for routing optimization.

2.3 Vehicle Routing and Rebalancing in E-Bike Systems

The operational component of our study extends the classical Capacitated Vehicle Routing Problem (CVRP). The foundational framework for the Static Bike Repositioning Problem (SBRP) was formalized by [15], who modeled routing and loading/unloading operations to minimize user dissatisfaction. [16] integrated inventory management with routing, determining service level requirements to prevent stock-outs. Specific to e-bikes, [22] suggests that when considering the dynamic process, the number of e-bikes and the remaining battery power level are the core factors to examine.

Our routing model diverges from strict cost-minimization; instead, we maximize an explicitly defined utility function $U_j(u_j, y_j)$ that evaluates the effective fulfillment of user demand within a maximum time limit T . As surveyed by [17], maximizing collected scores subject to a time budget aligns our model with the Orienteering Problem (OP) and Profitable Tour Problem (PTP) paradigms; a more recent survey can be found in [11]. To enforce chronological path progression and eliminate disconnected subtours, we utilize the Miller-Tucker-Zemlin (MTZ) constraints. As refined by [6], while MTZ formulations may possess weaker LP relaxations than fractional cut-generation techniques, their polynomial complexity makes them highly versatile—crucially, the MTZ structure allows effortless integration of node-specific, variable service times directly into the temporal tracking variable u_j , which is indispensable for calculating the time-dependent energy decay intrinsic to our utility function.

2.4 The “Predict-then-Optimize” Paradigm

Finally, the cohesive synthesis of machine learning predictions and mathematical programming reflects the emerging “Predict-then-Optimize” (PtO) paradigm. [3] provided a comprehensive methodological overview of integrating machine learning into combinatorial optimization. [7] introduced the Smart “Predict, then Optimize” (SPO) framework, arguing that predictive models should be evaluated based on the decision-making loss they ultimately produce. [19] further advanced this by proposing decision-focused learning via differentiable optimization layers.

Our methodology embodies this integrated philosophy. By explicitly feeding stochastic predictions into a dynamic utility function—one that penalizes blind supply via the Stock-out Risk Factor and evaluates actual energy conversion rates—our model ensures that the generated routing schedule is not merely a naive reaction to point predictions, but a robust strategic intervention optimized for the physical realities of the urban micro-mobility network.

3 Problem Description and Core Assumptions

3.1 Problem Context and Objective

We consider a shared e-bike service system partitioned into a large number of grids, with all e-bikes distributed across these grids. User behavior is highly stochastic: within any grid, users arrive randomly, select a bike at random for a trip, and then randomly return the bike to any grid, including the origin grid. If a user arrives at a grid while no serviceable bikes are available, the request is lost, degrading user experience and imposing a loss on the operator.

Within the service area, a service worker drives a vehicle equipped with a battery-swapping function and travels among the grids to perform battery replacement operations. All spare batteries are stored in a fixed charging cabinet; the service worker can carry only a limited number of batteries during each tour. The worker must periodically return to the charging cabinet to recharge depleted batteries and replenish inventory. Our goal is to develop a dynamic dispatch plan over a planning horizon of length T that specifies, for each grid, whether the worker should visit and how many batteries should be swapped.

In reality, the battery level distribution of shared e-bikes is continuous, which makes exact computation highly complex and requires tracking each bike individually—an approach that is undesirable in large-scale urban planning. Therefore, we classify e-bikes into three battery states: low (remaining battery below 20%), soon (remaining battery between 20% and 50%), and normal. Bikes in the low state are treated as unavailable. At each information update epoch (every T), we observe how many low-, soon-, and normal-state bikes are present at each grid. Bike states only change at information update epochs; between two updates, the state of each bike remains unchanged. For this system, our objective is to maximize the total utility from battery swapping, which is equivalent to minimizing operating cost consisting of fixed depreciation cost and penalty cost associated with lost user demand.

This process can be described as a closed-loop, optional-node, capacity-constrained vehicle routing problem. Compared with standard VRP formulations, its distinctive features are a structurally complex objective function and highly stochastic system dynamics. To abstract the system while preserving its essential stochastic character, we adopt the following assumptions.

3.2 Core Stochastic Assumptions

Assumption 1 (Poisson arrival and departure assumption): The arrival and departure processes of e-bikes at each grid $j \in \mathcal{J}$ follow independent Poisson processes with rates ρ_j and λ_j , respectively (the departure rate λ_j serves as a proxy for user demand):

$$N_{\text{in}}(t) \sim \text{Poisson}(\rho_j t), \quad N_{\text{out}}(t) \sim \text{Poisson}(\lambda_j t) \quad (3.1)$$

Assumption 2 (Stochastic user behavior assumption): For any bike currently in use anywhere in the service area, it may be returned to any grid. Therefore, for any grid, the battery-state distribution of arriving bikes can be regarded as the global battery-state distribution

Θ^{global} , with Θ_k^{global} denoting the proportion of bikes in state k where $k \in \{\text{low}, \text{soon}, \text{normal}\}$. Meanwhile, users randomly select one bike from the set of available bikes at the origin station. We denote by $\Theta_k^{\text{grid}}(t)$ the proportion of bikes in state k at time t within a given grid.

Assumption 3 (Fluid and expectation approximation assumption): For this system, using queueing-theoretic results would further increase nonlinearity and make the system intractable. We therefore treat the number of bikes as a continuous fluid, use rates in place of individual events, and replace random variables with their expectations. Under this approximation, the system dynamics become a continuously differentiable system, allowing us to obtain analytical solutions and perform optimization.

For an arbitrary grid j , with initial bike counts $n_{j,k}$ for each state k at $t = 0$, and with arrival rate ρ_j and departure rate λ_j , the inventory evolution of serviceable bikes is governed by:

$$N_{j,k}(t) = n_{j,k} + \int_0^t \left[\rho_j \cdot \Theta_k^{\text{global}} - \lambda_j \Theta_{j,k}^{\text{grid}}(\tau) \right] d\tau \quad (3.2)$$

Because of user stochasticity, the state distribution of arriving bikes is approximated by the same global distribution across all grids. For each fixed grid j , the local ODE has its own initial condition $n_{j,k}$ and rates ρ_j, λ_j , while sharing the common global proportions Θ_k^{global} . The local proportion is defined as

$$\Theta_{j,k}^{\text{grid}}(t) = \frac{N_{j,k}(t)}{N_j(t)}, \quad N_j(t) = \sum_{k=1}^2 N_{j,k}(t) \quad (3.3)$$

In this fluid ODE we only track the serviceable states $k = 1, 2$ (soon and normal); the low state remains excluded and acts as a reservoir convertible to serviceable inventory by swapping. Summing the equations over k yields:

$$\begin{aligned} \sum_{k=1}^2 \frac{dN_{j,k}(t)}{dt} &= \sum_{k=1}^2 \left(\rho_j \Theta_k^{\text{global}} - \lambda_j \frac{N_{j,k}(t)}{N_j(t)} \right) \\ \frac{d}{dt} N_j(t) &= \rho_j \sum_{k=1}^2 \Theta_k^{\text{global}} - \lambda_j \frac{1}{N_j(t)} \sum_{k=1}^2 N_{j,k}(t) \end{aligned} \quad (3.4)$$

Using $\sum_{k=1}^2 \Theta_k^{\text{global}} = 1$ and $\sum_{k=1}^2 N_{j,k}(t) = N_j(t)$, this simplifies to a linear ODE for $N_j(t)$:

$$\frac{dN_j(t)}{dt} = \rho_j - \lambda_j \quad (3.5)$$

With initial condition $N_j(0) = N_{j,0} = \sum_{k=1}^2 n_{j,k}$, integration gives the closed-form evolution:

$$N_j(t) = N_{j,0} + (\rho_j - \lambda_j)t \quad (3.6)$$

Substituting this back into the per-state ODE produces a first-order linear nonhomogeneous ODE:

$$\frac{dN_{j,k}}{dt} + \frac{\lambda_j}{N_{j,0} + (\rho_j - \lambda_j)t} N_{j,k} = \rho_j \Theta_k^{\text{global}} \quad (3.7)$$

This equation is of the standard form $y' + P_j(t)y = Q_j(t)$ and can be solved by the integrating factor method. Two cases arise depending on whether $\rho_j \neq \lambda_j$ or $\rho_j = \lambda_j$.

If $\rho_j \neq \lambda_j$, the integrating factor is

$$I_j(t) = \exp\left(\int \frac{\lambda_j}{N_{j,0} + (\rho_j - \lambda_j)t} dt\right) = (N_{j,0} + (\rho_j - \lambda_j)t)^{\frac{\lambda_j}{\rho_j - \lambda_j}} \quad (3.8)$$

and after integration one obtains

$$N_{j,k}(t) = \left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0}\right) \left(\frac{N_{j,0}}{N_{j,0} + (\rho_j - \lambda_j)t}\right)^{\frac{\lambda_j}{\rho_j - \lambda_j}} + \Theta_k^{\text{global}} (N_{j,0} + (\rho_j - \lambda_j)t) \quad (3.9)$$

If $\rho_j = \lambda_j$, the total serviceable inventory is conserved $N_j(t) \equiv N_{j,0}$ and the ODE reduces to a constant-coefficient linear equation with solution

$$N_{j,k}(t) = \Theta_k^{\text{global}} N_{j,0} + \left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0}\right) e^{-\frac{\lambda_j}{N_{j,0}} t} \quad (3.10)$$

In summary, the exact analytical solution of the per-grid ODE system is:

- If $\rho_j \neq \lambda_j$ (non-conserved per-grid serviceable inventory):

$$N_{j,k}(t) = \underbrace{\left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0}\right) \left(\frac{N_{j,0}}{N_{j,0} + (\rho_j - \lambda_j)t}\right)^{\frac{\lambda_j}{\rho_j - \lambda_j}}}_{\text{evolution of the initial deviation}} + \underbrace{\Theta_k^{\text{global}} (N_{j,0} + (\rho_j - \lambda_j)t)}_{\text{evolution of the theoretical quota}} \quad (3.11)$$

- If $\rho_j = \lambda_j$ (conserved per-grid serviceable inventory):

$$N_{j,k}(t) = \underbrace{\Theta_k^{\text{global}} N_{j,0}}_{\text{steady-state distribution}} + \underbrace{\left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0}\right) e^{-\frac{\lambda_j}{N_{j,0}} t}}_{\text{exponentially decaying transient term}} \quad (3.12)$$

In physical terms, the first part of each solution represents the deviation between the initial per-grid state and the ideal global quota. As time evolves this deviation term decays (or is diluted) and the per-grid composition converges toward the global proportions Θ_k^{global} .

4 系统动态与效用评估

4.1 自然状态下的库存动态

在无换电干预的条件下, 系统动态完全由用户需求驱动。任意网格内的状态分布 $\Theta_{j,k}^{\text{grid}}(t) = N_{j,k}(t)/N_j(t)$ 随时间逐渐收敛于全局分布 Θ_k^{global} , 而总库存 $N(t)$ 的演化取决于 ρ 与 λ 的关系: 当 $\lambda > \rho$ 时, $N(t)$ 逐渐下降并可能最终归零 (供给短缺); 当 $\lambda < \rho$ 时, 总库存持续增长 (可能出现过剩); 当 $\lambda = \rho$ 时, 总库存指数收敛至稳态, 稳态值与全局分布 Θ_k^{global} 相关。

在自然动态下, 对于每个网格 j , 可用服务车辆数量为

$$N_{\text{serv},j}(t) = N_{j,\text{soon}}(t) + N_{j,\text{normal}}(t) \quad (4.1)$$

在该流体模型中, 当可用库存归零时将触发需求损失。记零库存状态的起始时刻为 t_0 (其满足不动点方程 $N_{\text{serv},j}(t_0) = 0$), 则在区间 $[t_0, T]$ 内的损失需求为

$$\text{LostDemand}_j = \int_{t_0}^T (\lambda_j - \rho_j) dt = (\lambda_j - \rho_j)(T - t_0) \quad (4.2)$$

对该系统而言, 任意节点的服务价值体现于对未来净需求的满足。从任意到达时刻 u_j 直至有效边界 $\min(t_0, T)$, 净服务价值可表达为质量加权的流出服务率与质量加权的流入稀释率之差:

$$S_j(u_j) = \int_{u_j}^{\min(t_0, T)} \left(\lambda_j (\Theta_{j, \text{normal}}^{\text{grid}}(t) + 0.8 \Theta_{j, \text{soon}}^{\text{grid}}(t)) - \rho_j (\Theta_{\text{normal}}^{\text{global}} + 0.8 \Theta_{\text{soon}}^{\text{global}}) \right) dt \quad (4.3)$$

4.2 瞬时换电后的库存动态

假设在时刻 u , 服务工人到达网格 j 并执行换电操作。其操作规则如下:

- 换电仅适用于 low 和 soon 状态的车辆。换电后, 车辆被重置为 normal 状态并恢复完全可用性。
- 不同电池状态对应不同的服务质量: normal 为最优 (100% 服务), low 为最差 (0% 服务), soon 为中等 (80% 服务水平)。
- 服务工人的车载电池数量有限, 因此每次访问的换电量受到约束。工人能够识别网格内所有车辆的状态, 并优先更换 low 状态车辆, 若有剩余容量再更换 soon 状态车辆。

在上述假设下, 若服务工人在时刻 u 执行 y 次换电, 换电后的库存仍遵循相同的 ODE 结构, 但初始条件发生了改变。换电后的总库存与各状态存量如下:

$$\tilde{N}_{j,0} = N_j(u) + \min\{y_j, N_{j,\text{low}}(u)\} \quad (4.4)$$

$$\tilde{n}_{j,k} = \begin{cases} N_{j,k}(u) - \min\{y_j, N_{j,\text{low}}(u)\} & \text{if } k = \text{low} \\ N_{j,k}(u) - \max\{0, y_j - N_{j,\text{low}}(u)\} & \text{if } k = \text{soon} \\ N_{j,k}(u) + y_j & \text{if } k = \text{normal} \end{cases} \quad (4.5)$$

换电后的库存 $\tilde{N}_{j,k}(t)$ ($t > u$) 遵循第 3 节推导的相同解析形式, 仅需将初始条件 $n_{j,k}$ 替换为 $\tilde{n}_{j,k}$, 并将时间原点平移至 u :

- 若 $\rho \neq \lambda$:

$$\tilde{N}_{j,k}(t) = \left(\tilde{n}_{j,k} - \Theta_k^{\text{global}} \tilde{N}_{j,0} \right) \left(\frac{\tilde{N}_{j,0}}{\tilde{N}_{j,0} + (\rho - \lambda)(t - u)} \right)^{\frac{\lambda}{\rho - \lambda}} + \Theta_k^{\text{global}} (\tilde{N}_{j,0} + (\rho - \lambda)(t - u)) \quad (4.6)$$

- 若 $\rho = \lambda$:

$$\tilde{N}_{j,k}(t) = \Theta_k^{\text{global}} \tilde{N}_{j,0} + \left(\tilde{n}_{j,k} - \Theta_k^{\text{global}} \tilde{N}_{j,0} \right) e^{-\frac{\lambda}{\tilde{N}_{j,0}}(t - u)} \quad (4.7)$$

换电服务的本质在于通过改变 ODE 的初始条件来调整系统的动态演化轨迹, 从而达到延迟缺货的目的。记 t_0 为未换电时的原始缺货点, $\tilde{t}_0$ 为换电后的新缺货点。整个规划周期 T 以换电时刻 u 为界被分为两段: 在 $t \leq u$ 区间内, 系统按自然动态演化; 在 $t > u$ 区间内, 换电干预发生作用。

然而, 由于 soon 状态与 normal 状态的车辆所提供的可骑行距离存在显著差异, 二者不能被视为提供同等服务水平的资源。定义 $\tilde{\Theta}_{j,k}^{\text{grid}}(t) = \tilde{N}_{j,k}(t)/\tilde{N}_j(t)$ 以追踪换电后的状态分布。假

定 20% 的用户因车辆电量有限而无法完成完整订单，我们对 soon 状态赋予 80% 的服务权重。换电后的服务价值为：

$$\tilde{S}_j(u_j, y_j) = \int_{u_j}^T \left(\lambda_j (\tilde{\Theta}_{j,\text{normal}}^{\text{grid}}(t) + 0.8 \tilde{\Theta}_{j,\text{soon}}^{\text{grid}}(t)) - \rho_j (\Theta_{\text{normal}}^{\text{global}} + 0.8 \Theta_{\text{soon}}^{\text{global}}) \right) dt \quad (4.8)$$

换电操作的边际效用即为服务价值之差：

$$U_j(u_j, y_j) = \tilde{S}_j(u_j, y_j) - S_j(u_j) \quad (4.9)$$

4.3 质量调整后的封闭效用函数

为了在不引入高维随机变量的前提下精确刻画 normal 与 soon 状态之间因用户”里程焦虑”产生的服务水平差异，我们基于瞬时期望质量定义了一个连续的服务评估框架。由于用户随机选择可用车辆，在时刻 t 选中状态 k 车辆的概率正比于其瞬时库存比例 $\Theta_{j,k}^{\text{grid}}(t) = N_{j,k}(t)/N_j(t)$ 。我们采用以下约定：

- 在自然动态下，low 状态车辆不参与可用库存的 ODE 演化，仅作为在换电瞬间可转化的静态库存池。
- 若到达时间 $u_j \geq \tau_0$ （或换电后 $u_j \geq \tilde{\tau}_0$ ），则该节点在剩余周期内不提供净服务价值。
- 运营效用 $U_j(u_j, y_j)$ 在优化中被截断为非负值。

在 soon 状态存在 20% 服务降级的前提下，从到达时刻 u_j 至有效边界 $\tau_0 = \min(t_0, T)$ 所收集的基线净服务价值为：

$$S_j(u_j) = \int_{u_j}^{\tau_0} \left[\lambda_j \left(1.0 \cdot \frac{N_{j,\text{normal}}(t)}{N_j(t)} + 0.8 \cdot \frac{N_{j,\text{soon}}(t)}{N_j(t)} \right) - \rho_j \left(1.0 \cdot \Theta_{\text{normal}}^{\text{global}} + 0.8 \cdot \Theta_{\text{soon}}^{\text{global}} \right) \right] dt \quad (4.10)$$

若直接对此积分进行数值求解，其计算代价极高。然而，我们可以利用 ODE 系统的一个结构性特征来避免数值积分。将第 3 节中 $k = \text{normal}$ 和 $k = \text{soon}$ 的控制方程分别乘以对应的质量权重（1.0 和 0.8）并线性相加：

$$\begin{aligned} \frac{d}{dt} (N_{j,\text{normal}}(t) + 0.8 N_{j,\text{soon}}(t)) &= \rho_j \left(\Theta_{\text{normal}}^{\text{global}} + 0.8 \Theta_{\text{soon}}^{\text{global}} \right) \\ &\quad - \frac{\lambda_j}{N_j(t)} (N_{j,\text{normal}}(t) + 0.8 N_{j,\text{soon}}(t)) \end{aligned} \quad (4.11)$$

将此式与被积函数对比，可发现一个关键洞察：系统净服务效用的流出率恰好等于系统内部”质量加权库存”总导数的相反数。根据微积分基本定理，该积分直接化简为边界求值：

$$S_j(u_j) = [N_{j,\text{normal}}(u_j) + 0.8 N_{j,\text{soon}}(u_j)] - [N_{j,\text{normal}}(\tau_0) + 0.8 N_{j,\text{soon}}(\tau_0)] \quad (4.12)$$

在时刻 u_j 完成 y_j 块电池的换电后，库存瞬间跃升为 $\tilde{n}_{j,k}$ 。换电后的服务价值遵循完全相同的数学结构：

$$\tilde{S}_j(u_j, y_j) = [\tilde{n}_{j,\text{normal}} + 0.8 \tilde{n}_{j,\text{soon}}] - [\tilde{N}_{j,\text{normal}}(\tilde{\tau}_0) + 0.8 \tilde{N}_{j,\text{soon}}(\tilde{\tau}_0)] \quad (4.13)$$

其中 $\tilde{\tau}_0 = \min(\tilde{t}_0, T)$ 为换电所延缓的新的有效服务上限。

净运营效用 $U_j(u_j, y_j)$ 为换电前后服务价值的边际增益：

$$U_j(u_j, y_j) = \tilde{S}_j(u_j, y_j) - S_j(u_j) = \Delta_{\text{swap}} - \left[\left(\tilde{N}_{j,\text{normal}}(\tilde{\tau}_0) + 0.8\tilde{N}_{j,\text{soon}}(\tilde{\tau}_0) \right) - (N_{j,\text{normal}}(\tau_0) + 0.8N_{j,\text{soon}}(\tau_0)) \right] \quad (4.14)$$

其中 Δ_{swap} 表示在换电瞬间注入系统的净服务质量增量：

$$\Delta_{\text{swap}} = \tilde{n}_{j,\text{normal}} - N_{j,\text{normal}}(u_j) + 0.8(\tilde{n}_{j,\text{soon}} - N_{j,\text{soon}}(u_j)) \quad (4.15)$$

结合优先换电逻辑（先 low 后 soon）， Δ_{swap} 可显式表达为 y_j 的极简函数：

$$\Delta_{\text{swap}} = \min\{y_j, N_{j,\text{low}}(u_j)\} + 0.2 \max\{0, y_j - N_{j,\text{low}}(u_j)\} \quad (4.16)$$

该闭式解提供了理论严谨且计算极轻量的评估原语：将复杂的连续时间随机积分降维为常数时间复杂度的代数运算，使效用函数能够直接内嵌于大规模 MIP 路由优化之中。

5 全局路由优化

建立了任意节点的效用评价体系后，调度问题自然转化为：在车载电池容量 C 和规划周期 T 的双重约束下，选择访问哪些网格、以何种顺序访问、在每个网格换多少块电池，使得全路径的效用总和最大。这是一个可选节点、容量受限、时间受限的车辆路径问题，其核心难点在于目标函数的非线性——效用 $U_j(u_j, y_j)$ 通过 ODE 导出的闭式解同时依赖于到达时刻 u_j 和换电量 y_j 。

令 $\mathcal{N} = \mathcal{J} \cup \{0\}$ ，其中 0 为车场。决策变量包括： $x_{ij} \in \{0, 1\}$ （路由）、 $v_j \in \{0, 1\}$ （访问）、 $y_j \in \mathbb{Z}^+$ （换电量）和 $u_j \geq 0$ （到达时间）。记 c 为单块电池的换电耗时， a_{ij} 为行驶时间。基础 MIP 模型如下：

$$\max \sum_{j \in \mathcal{J}} \sum_{y \in \mathcal{Y}} \sum_{s \in \mathcal{S}} \Omega_j(y, \tau_s) \cdot \lambda_{j,y,s} \quad (5.1)$$

$$\text{s.t.} \quad \sum_{i \in \delta^-(j)} x_{ij} = v_j, \quad \sum_{k \in \delta^+(j)} x_{jk} = v_j, \quad \forall j \in \mathcal{J} \quad (5.2)$$

$$\sum_{j \in \mathcal{J}} x_{0j} = \sum_{i \in \mathcal{J}} x_{i0} \leq 1 \quad (5.3)$$

$$y_j \leq C \cdot v_j, \quad \sum_{j \in \mathcal{J}} y_j \leq C \quad (5.4)$$

$$u_j \geq u_i + c \cdot y_i + a_{ij} - M(1 - x_{ij}), \quad \forall i \in \mathcal{N}, j \in \mathcal{J}, i \neq j \quad (5.5)$$

$$u_i + c \cdot y_i + a_{i0} - M(1 - x_{i0}) \leq T, \quad \forall i \in \mathcal{J} \quad (5.6)$$

$$u_j = \sum_{y \in \mathcal{Y}} \sum_{s \in \mathcal{S}} \tau_s \cdot \lambda_{j,y,s}, \quad \forall j \in \mathcal{J} \quad (5.7)$$

$$x_{ij}, v_j \in \{0, 1\}, \quad y_j \in \mathbb{Z}^+, \quad u_j \geq 0 \quad (5.8)$$

其中 $\Omega_j(y, \tau_s)$ 为预计算的效用张量——表示在离散时刻 τ_s 到达网格 j 并换电 y 块的边际效用； $\lambda_{j,y,s}$ 为 SOS2 插值权重变量，用于将连续到达时间映射至离散断点； $\mathcal{Y}$ 和 $\mathcal{S}$ 分别为离散换电量和时间断点索引集合。

约束 (5.2) 保证流守恒, (5.3) 保证车场出发与返回一致性, (5.4) 为换电容量约束, (5.5) 为 MTZ 子回路消除与时间推进约束, (5.6) 为规划周期截止约束。

5.1 MTZ Big-M 约束的结构性缺陷

该模型的核心计算困难源于约束 (5.5)。这是一个典型的 Big-M 约束：当 x_{ij} 在 LP 松弛中取分数值时，松弛项 $M(1 - x_{ij})$ 使得到达时间 u_j 与 u_i 完全解耦。求解器可以通过在大量弧段上分配微小分数流，在满足所有流守恒约束的同时构造一个时间推进完全虚构的伪可行解。这导致根节点 LP 松弛产生极为松散的上界——通常高达真实整数最优解的 2-3 倍——使分支定界几乎无法有效剪枝。

我们先后开发并测试了三种逐步增强的 MIP 建模架构：

方案 A：SOS2-PLA + 全量 MTZ。最直接的实现方式，使用 SOS2 约束对效用函数进行分段线性插值，搭配完整的 $O(N^2)$ 条显式 MTZ 约束。在 490 个活跃网格、 $P = 5$ 段离散化的设定下，该方案在 1800s 后超时截断，gap 高达 92.6%。根节点 LP 松弛目标值为 2.67，而最佳整数解仅为 1.35。

方案 B：Delta 增量编码 + MCF 流约束 + 延迟 MTZ 回调。该方案做了三点改进：(1) 将 SOS2 替换为二元 delta 阶梯变量 ($\delta_1 \geq \delta_2 \geq \dots \geq \delta_P$)，消除了 SOS2 的特殊结构开销并精确刻画了到达时间的凸组合性质；(2) 引入单商品流 (MCF) 约束—— $f_{ij} \leq C \cdot x_{ij}$ 以及每个节点的电池流量守恒——加强路由变量之间的 LP 耦合强度；(3) 将 $O(N^2)$ 条 MTZ 约束从显式行转化为延迟约束回调 (MIPSOL)，仅在发现整数可行解时才注入违规的时序割。该方案将 1800s 后的 gap 降至 31.1%，目标函数值提升至 1.998。然而，LazyConstraints 的启用迫使 Gurobi 禁用激进预求解和割平面生成，限制了后续进展。

方案 C：时空图网络流模型 (Space-Time Graph, STGraph)。该方案从根本上消灭了所有 Big-M 约束。核心思想是将连续时间显式离散化并嵌入有向无环图 (DAG) 的拓扑结构中——时间步成为图的固有维度，有向边本身就蕴含了时间推进的方向性。在此框架下，MTZ 约束不再需要（时间不可逆性保证了子回路在结构上不可能出现），且所有时间同步约束退化为网络流平衡方程。虽然变量数量大幅增加（约 268 万列 vs. 方案 B 的 32.7 万列），但 LP 松弛极其紧致：根节点 LP 目标值 1.034 与最优整数解 1.031 的偏差不足 0.3%。求解器在 126 秒内即证得全局最优（gap 0.26%），仅探索了 1 个 B&B 节点。

表 1: 三种优化架构的求解性能对比（490 个活跃网格， $C = 20$ ， $T = 1.0h$ ，时限 1800s）

方案	变量数	约束数	根节点 LP	最优目标值	Gap	耗时
A: SOS2+MTZ	171k	144k	2.670	1.348	92.6%	1800s (超时)
B: Delta+MCF+Lazy	327k	232k	2.636	1.998	31.1%	1800s (超时)
C: STGraph	2678k	8.3k	1.034	1.031	0.26%	126s (最优)

STGraph 架构的压倒性优势揭示了整数规划中的一个经典洞察：**约束矩阵的紧致性远比模型规模重要**。STGraph 虽然变量数量最大，但其约束矩阵本质上是一个网络流矩阵（具有全幺模子结构）外加少量容量和链接约束，因此 LP 松弛自然紧致。相反，方案 A 和 B 的约束矩阵中嵌入了大量 Big-M 项，导致 LP 松弛与整数包络之间存在严重的“断层”——这个断层需要海量的分支定界搜索来弥合。

6 模型求解：时空图网络流优化框架

6.1 求解管线总览

完整的求解管线包含六个阶段：

1. **数据加载与快照提取。**从预测数据集中提取目标时刻的全区域快照，包括每个 H3 六边形网格的 low/soon/normal 车辆数量、预测租还速率 $\hat{\lambda}_j, \hat{\rho}_j$ ，以及网格中心坐标。
2. **旅行时间矩阵构建。**基于 Haversine 球面距离公式，以恒定车速 30 km/h 计算所有网格对之间的行驶时间。车场位置取所有活跃网格的几何质心。
3. **静态效用张量预计算 (PLA)。**在 P 个等距时间断点 $\{\tau_0 = 0, \tau_1, \dots, \tau_P = T\}$ 上，对每个网格 j 的每个可行换电量 $y \in \mathcal{Y}$ ，调用第 4 节的封闭效用函数 $U_j(y, \tau_s)$ ，构成三维效用张量 $\Omega_j(y, \tau_s)$ 。该张量是下游 MIP 模型中目标函数系数的唯一来源。
4. **Geo-Fencing 空间剪枝。**剔除在所有 (y, s) 组合下效用恒为零的网格。对剩余网格应用 K 近邻 (KNN) 空间稀疏化：每个节点仅保留行驶时间最短的 $K = 50$ 个邻居作为候选弧段，压缩决策空间约 90%，同时保留最具物理意义的近邻转移。
5. **时空图构建与 MIP 求解。**基于 KNN 空间网络和时间离散化构建时空扩展图，声明网络流决策变量与辅助联动变量，施加流守恒、容量、单次访问和联动约束，注入贪心热启动初始解，调用 Gurobi 进行单阶段 branch-and-cut 求解。
6. **路由解码与时序审计。**沿 DAG 拓扑追踪激活的时空弧，还原为一维物理路由序列。基于真实的 Haversine 几何进行时序一致性的双向交叉校验。

6.2 时空图 (STGraph) 构建方法

STGraph 的核心思想是将“时间”从约束的右侧提升为图的显式维度。令 $\mathcal{T} = \{0, 1, \dots, P\}$ 为离散时间步索引集合，每个时间步 $s \in \mathcal{T}$ 对应物理时刻 τ_s 。时空图中的节点为二元组 (i, s) ，表示“车辆在时刻 τ_s 位于节点 i ”。时空图中的有向弧为五元组 $e = (i, s_1, j, s_2, y)$ ，表示“车辆在时刻 τ_{s_1} 从节点 i 出发，在 i 处换电 y 块，行驶 a_{ij} 时间后，在时刻 τ_{s_2} 到达节点 j ”。

时间右取整对齐。由于物理到达时间很少恰好落在离散断点上，我们采用右取整 (right-rounding) 策略：

$$s_2 = \min\{s \in \mathcal{T} : \tau_s \geq \tau_{s_1} + c \cdot y + a_{ij}\} \quad (6.1)$$

这确保了所有非车场弧满足 $s_2 \geq s_1 + 1$ ，使时空图天然构成 DAG——时间单调性成为一种拓扑性质而非需要施加的约束。右取整引入的保守时间松弛最多为一个离散间隔的宽度，其大小可由时间分辨率 P 控制。

KNN 空间网络稀疏化。对于 $N = 490$ 个活跃网格，全连接空间网络包含 $N(N - 1) \approx 240,000$ 条弧。乘以 P 个时间步和 $|\mathcal{Y}|$ 种换电选择后，决策变量数量将达数亿。KNN 稀疏化 ($K = 50$) 结合时间步可行性过滤，将实际弧段压缩至约 270 万条——缩减约两个数量级。

时空弧枚举。时空弧分为两类：

- **首发弧：** $(0, 0) \rightarrow (j, s_2)$, $y = 0$ ，对所有从车场出发的行驶时间在 T 内的网格 j 。

- **转移弧**: $(i, s_1) \rightarrow (j, s_2)$, $y \in \mathcal{Y}$, 对所有 $s_1 < P$, $j \in \text{KNN}(i)$, 且右取整到达步 $s_2 \leq P$ 。

6.3 STGraph-MIP 数学模型

令 $\mathcal{E}$ 为所有时空弧的集合, 对每条弧 $e = (i^e, s_1^e, j^e, s_2^e, y^e)$ 定义二元决策变量 $x_e \in \{0, 1\}$ 。辅助联动变量 v_j, u_j, y_j 用于与传统 VRP 结果解析管线兼容。

6.3.1 目标函数

$$\max \sum_{e \in \mathcal{E}: i^e \neq 0} \Omega_{i^e}(y^e, \tau_{s_1^e}) \cdot x_e \quad (6.2)$$

每条弧的目标系数直接从预计算的效用张量中按弧的出发节点、换电量和出发时间步查表取值。效用语义与第 4 节一致：效用由到达时间和换电量决定；在离散 STGraph 中，到达时间 u_j 由离散步 τ_{s_1} 近似。

6.3.2 时空流守恒约束

对每个网格 j 和每个时间步 s , 流入等于流出：

$$\sum_{e \in \delta_{ST}^-(j, s)} x_e = \sum_{e \in \delta_{ST}^+(j, s)} x_e, \quad \forall j \in \mathcal{J}, s \in \mathcal{T} \quad (6.3)$$

其中 $\delta_{ST}^-(j, s)$ 和 $\delta_{ST}^+(j, s)$ 分别为时空节点 (j, s) 的入弧和出弧集合。这是标准网络流平衡约束。关键在于，这一组约束一次性替代了传统模型中所有的 MTZ 子回路消除、时间推进和时间同步约束——在 DAG 中，流守恒本身就保证了物理可行性。

6.3.3 车场边界与单次访问约束

$$\sum_{e \in \delta_{ST}^+(0, 0)} x_e \leq 1 \quad (6.4)$$

$$\sum_{s=1}^P \sum_{e \in \delta_{ST}^-(0, s)} x_e = \sum_{e \in \delta_{ST}^+(0, 0)} x_e \quad (6.5)$$

$$\sum_{s_1=0}^{P-1} \sum_{e \in \delta_{ST}^+(j, s_1)} x_e \leq 1, \quad \forall j \in \mathcal{J} \quad (6.6)$$

约束 (6.4) 限制至多一辆服务车从车场出发。约束 (6.5) 确保出发的车辆必须在时间步 1 至 P 之间的某一步返回车场。约束 (6.6) 保证每个物理网格最多被访问一次，杜绝车辆在不同时间步反复访问同一网格以重复获取效用。

6.3.4 全局容量约束

$$\sum_{e \in \mathcal{E}} y^e \cdot x_e \leq C \quad (6.7)$$

全程累积换电总量不超过车载电池容量 C 。

6.3.5 联动约束

为使优化结果能够与标准 VRP 结果解析管线无缝衔接，我们建立了时空弧变量向传统 VRP 变量的显式映射：

$$v_j = \sum_{s_1=0}^{P-1} \sum_{e \in \delta_{ST}^+(j, s_1)} x_e, \quad \forall j \in \mathcal{J} \quad (6.8)$$

$$u_j = \sum_{s_2=0}^P \tau_{s_2} \sum_{e \in \delta_{ST}^-(j, s_2)} x_e, \quad \forall j \in \mathcal{J} \quad (6.9)$$

$$y_j = \sum_{s_1=0}^{P-1} \sum_{e \in \delta_{ST}^+(j, s_1)} y^e \cdot x_e, \quad \forall j \in \mathcal{J} \quad (6.10)$$

约束 (6.8) 将访问标志等价于是否存在从 j 出发的时空弧。约束 (6.9) 将到达时间锚定为进入时空节点 (j, s_2) 所对应的物理时间 τ_{s_2} 。约束 (6.10) 将换电量映射为激活出弧上携带的 y^e 值。

6.3.6 变量域

$$x_e \in \{0, 1\}, \forall e \in \mathcal{E}; \quad v_j \in \{0, 1\}, u_j \geq 0, y_j \in \{0, 1, \dots, C\}, \forall j \in \mathcal{J} \quad (6.11)$$

6.4 贪心热启动启发式

为确保 Gurobi 在根节点即获得高质量的初始可行解（primal bound），我们采用基于效用/行驶时间比的贪心最近邻构造启发式。从车场出发，对当前节点 KNN 邻域内的每个未访问网络 j ，评估其在预计到达时间步下所有可行换电量的最大效用，按效用/行驶时间比排序，选择最优者，并验证返程可行性（若无法在时间 T 内返回车场，则逐级递减换电量以腾出返程时间）。该启发式的复杂度为 $O(N \cdot K \cdot |\mathcal{Y}|)$ ，通常在 0.1 秒内完成。生成的物理动作序列被转化为时空弧激活模式，作为 Gurobi MIP Start 注入。由于启发式建立在与 MIP 模型完全相同的 KNN 图和时间离散化规则之上，所构造的路径在 STGraph 中天然可行。

6.5 求解器参数配置

Gurobi 参数的调优充分针对 STGraph 的结构特性：

- **根节点 LP 使用 Barrier 方法。** STGraph 的约束矩阵是一个大规模、稀疏、高度结构化的网络流矩阵，Barrier（内点法）的收敛速度远超单纯形法。实测中 32 次 Barrier 迭代（121s）即收敛至 10^{-12} 精度。

- **禁用 Crossover。**将 Barrier 基转换为单纯形可行基 (crossover) 在网络流矩阵上异常耗时且对后续 B&B 价值有限——直接使用 Barrier 基。
- **完整预求解与适度割平面。**与 Delta-MCF 模型不同, STGraph 无需 LazyConstraints, 因此 Gurobi 可以正常进行完整的代数预求解 (削减 51% 的变量和 44% 的约束) 和割平面生成。
- **分支优先级。**访问变量 v_j 被赋予最高分支优先级, 因为访问集合一旦确定, 换电量 y_j 和到达时间 u_j 即被时空流守恒唯一锁定。

6.6 求解结果与对比分析

在相同的输入数据 (651 个原始网格, 目标时刻 2025-10-23 12:00, $T = 1.0\text{h}$, $C = 20$, Intel i5-10210U @ 1.60GHz, 8 线程) 下, 三种方案的求解轨迹呈现出本质差异。

最优解质量。STGraph 以 126 秒求得目标值 1.031, 经 Gurobi 证明为全局最优 (gap 0.26%)。最优调度方案访问 11 个网格, 用尽全部 20 块电池, 行驶净耗时 0.462h, 换电净耗时 0.400h, 周期总长 0.862h。Delta-MCF 在 1800s 超时截断于目标值 1.998, 剩余 gap 31.1%, 访问 10 个网格但时序审计发现 10 处偏差超过 0.05h (延迟 MTZ 回调注入不完全导致的时间不一致)。SOS2 超时截断于 1.348, 剩余 gap 92.6%, 访问 12 个网格但 B&B 树已扩展至 3538 个节点而无收敛迹象。

根节点 LP 松弛质量。三者的根节点 LP 目标值分别为: STGraph 1.034, Delta-MCF 2.636, SOS2 2.670。STGraph 的根节点松弛与最优整数解的比值仅为 1.003 (偏离 0.3%), 意味着 LP 松弛几乎已经“知道”最优解的位置。Delta-MCF 和 SOS2 的对应比值分别为 1.319 和 1.981, 松弛解严重高估了可实现效用, 导致 B&B 下界长期无法提升。

变量规模与求解速度的“悖论”。STGraph 拥有 268 万个变量 (其中 267 万为二元变量)——是 Delta-MCF 的 8 倍、SOS2 的 16 倍——却比二者分别快 14 倍。这一看似矛盾的现象揭示了整数规划中的一条核心规律: **模型的紧致性远比问题规模更能决定求解效率。**STGraph 以大规模但结构良性 (网络流结构) 的约束矩阵换取了所有 Big-M 约束的完全消除。由此带来的界改进——从根节点 LP gap 的 30–100% 压缩至不足 0.5%——几乎消灭了所有分支搜索, 总求解时间随之坍缩。

预求解效果差异。STGraph 的预求解阶段削减了 51% 的变量 (从 268 万降至 52 万), 而 Delta-MCF 受 LazyConstraints=1 限制仅削减了 0.07%。充分的预求解能力是模型结构清晰度的直接反映: 不存在 Big-M 项和延迟约束回调的纯网络流模型天然更适合求解器的代数化简。

综上, STGraph 重构方案通过将时间耦合从 Big-M 不等式约束转化为 DAG 拓扑, 从根本上化解了本问题的核心矛盾——将复杂的时间依赖非线性效用嵌入车辆路径框架。这一方法论启示超越了换电路径优化的特例范畴: 任何目标函数或约束对连续时间有非平凡依赖的 VRP 变种, 都可能通过时空图重构从源头消除 Big-M 缺陷而获益。

参考文献

- [1] Deniz Akturk, Ozan Candogan, and Varun Gupta. Managing resources for shared micromobility: Approximate optimality in large-scale systems. *Management Science*, 71(7):5676–5695, 2024.
- [2] Michel Benaim and Jean-Yves Le Boudec. A class of mean field interaction models for computer and communication systems. *Performance Evaluation*, 65(11–12):823–838, 2008.
- [3] Yoshua Bengio, Andrea Lodi, and Antoine Prouvost. Machine learning for combinatorial optimization: A methodological tour d’horizon. *European Journal of Operational Research*, 290(2):405–421, 2021.
- [4] Anton Braverman, J. G. Dai, Xin Liu, and Lei Ying. Empty-car routing in ridesharing systems. *Operations Research*, 67(5):1437–1452, 2019.
- [5] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [6] Martin Desrochers and Gilbert Laporte. Improvements and extensions to the Miller-Tucker-Zemlin subtour elimination constraints. *Operations Research Letters*, 10(1):27–36, 1991.
- [7] Adam N. Elmachtoub and Paul Grigas. Smart ”predict, then optimize”. *Management Science*, 68(1):9–26, 2022.
- [8] Christine Fricker and Nicolas Gast. Incentives and redistribution in homogeneous bike-sharing systems with stations of finite capacity. *EURO Journal on Transportation and Logistics*, 5(3):261–291, 2016.
- [9] Nicolas Gast. Expected values estimated via mean-field approximation are $1/N$ -accurate. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, 1(1):1–26, 2017.
- [10] Xiucheng Geng, Yaguang Li, Lijun Wang, Lei Zhang, Qiang Yang, Jun Ye, and Yan Liu. Spatiotemporal multi-graph convolution network for ride-hailing demand forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 3656–3663, 2019.
- [11] Aldy Gunawan, Hoong Chuin Lau, and Pieter Vansteenwegen. Orienteering problem: A survey of recent variants, solution approaches and applications. *European Journal of Operational Research*, 255(2):315–332, 2016.
- [12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

- [13] Yaguang Li, Richard Rose, and Eitan Elnekave. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In *International Conference on Learning Representations*, 2018.
- [14] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. Catboost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [15] Tami Raviv, Michal Tzur, and Itai A. Forma. Static repositioning in a bike-sharing system: Models and solution approaches. *EURO Journal on Transportation and Logistics*, 2(3):187–229, 2013.
- [16] Jasper Schuijbroek, Robert C. Hampshire, and Willem-Jan Van Hoes. Inventory routing and pricing in bike-sharing systems. *European Journal of Operational Research*, 258(3):971–985, 2017.
- [17] Pieter Vansteenwegen, Wouter Souffriau, and Dirk Van Oudheusden. The orienteering problem: A survey. *European Journal of Operational Research*, 209(1):1–10, 2011.
- [18] Ariel Waserhole and Vincent Jost. Pricing in vehicle sharing systems: Optimization in queuing networks with product forms. *EURO Journal on Transportation and Logistics*, 5(3):293–320, 2016.
- [19] Bryan Wilder, Bistra Dilikina, and Milind Tambe. Melding the data-decisions pipeline: Decision-focused learning for combinatorial optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 1658–1665, 2019.
- [20] Bing Yu, Haoteng Yin, and Zhanxing Zhu. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence*, pages 3634–3640, 2018.
- [21] Chuanpan Zheng, Xiaoliang Fan, Cheng Wang, and Jianzhong Qi. GMAN: A graph multi-attention network for traffic prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 1234–1241, 2020.
- [22] Yaoming Zhou, Zeyu Lin, Rui Guan, and Jiuh-Bing Sheu. Dynamic battery swapping and rebalancing strategies for e-bike sharing systems. *Transportation Research Part B: Methodological*, 177:102820, 2023.